

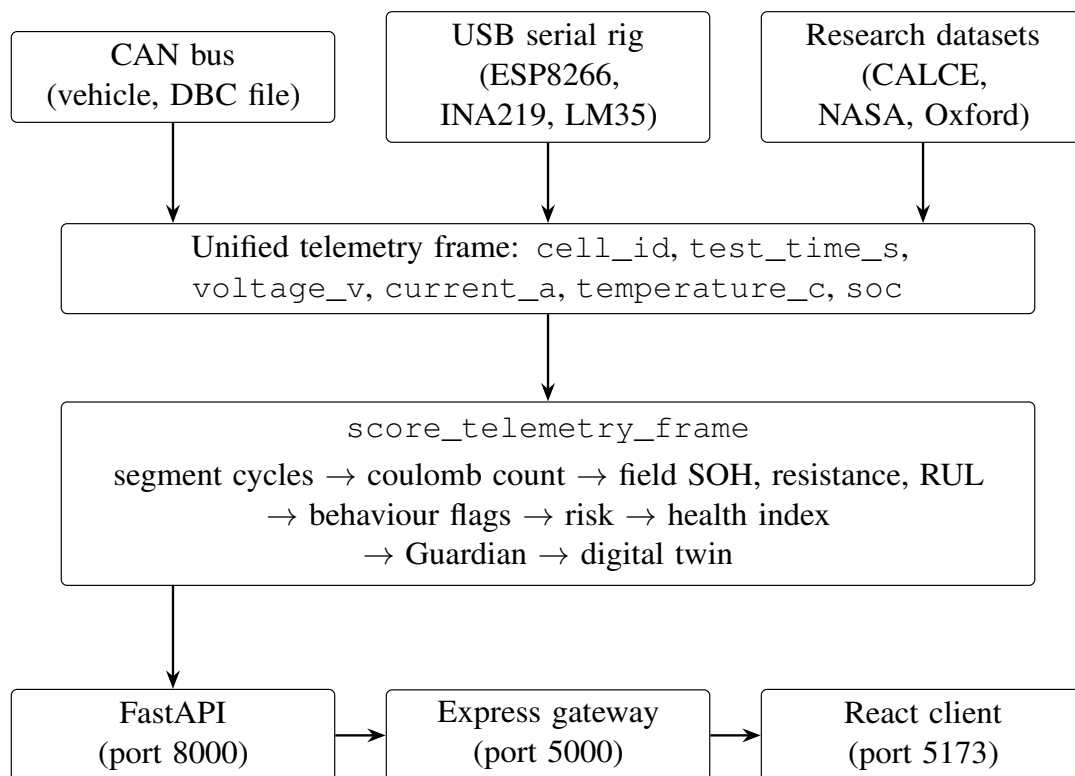


Figure 3.1: Overall system framework: three transports, one unified frame, one scoring function, and the service tier.

in output is a difference in the telemetry.

3.5 DATASETS

Four sources are used (Table 3.2). They were chosen because they differ in chemistry, format, temperature and protocol.

Table 3.2: Data Sources and Their Role

Source	Cells and conditions	Role	Ground truth used
CALCE CS2/CX2	22 prismatic LCO cells, room temperature, 7 protocols (constant-current, multi-rate, real partial cycling)	Development and primary validation	Cycler-measured discharge capacity
NASA PCoE	18650 cells, 9 protocols, ambient 4–44 °C; 31 cells in the benchmark, 27 in the cross-dataset study	Benchmark and third-dataset test	NASA’s computed capacity
Oxford Dataset 1	8 Kokam NMC/LCO pouch cells, 740 mAh, 40 °C, 519 characterisations	External validation, unchanged	1C capacity at each characterisation
Bench rig	1 HONGLI 18650 cell, at rest in the two real captures (83 and 85 records)	Data acquisition only	Frame integrity and timing

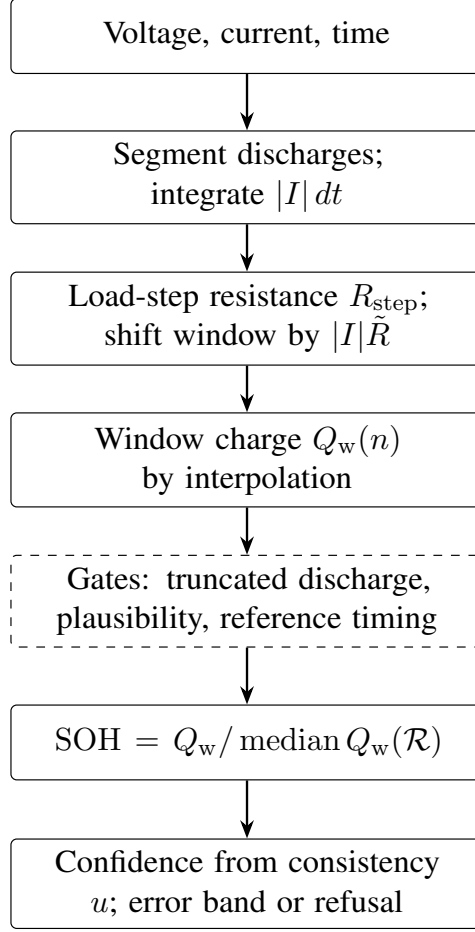


Figure 3.2: Field SOH estimator: from raw telemetry to a banded or refused output.

is shifted to

$$[V_{\text{low}} - |I_n| \tilde{R}(n), V_{\text{high}} - |I_n| \tilde{R}(n)]. \quad (3.3)$$

Discharges preceding the first resistance estimate are left unmeasured rather than scored uncompensated.

3.11 DATA-QUALITY GATES AND REFUSAL CONDITIONS

Each gate needs no ground truth and can therefore run in deployment.

1. **Truncated discharge.** A discharge delivering less than 50% of the reference discharges' total charge is not scored. A cut-short cycle crosses the window with little charge and would read as severe fade.
2. **Physical plausibility.** A ratio below 0.50 is withheld, and a cell with more than 20% of readings below it is refused entirely. A ratio above 1.10 is withheld as not spanning the same part of the curve.
3. **Reference timing.** The reference must be complete within 20 equivalent full cycles of charge throughput. Throughput rather than a discharge count is used because storage-protocol records contain thousands of short pulses.

rated capacity), D (data) and S (status).

```
BEACON1 D {"t":12.5,"v":3.91,"i":-1.48,"tc":27.4,"soc":82.1}*06
```

Figure 4.1: A data frame: sentinel BEACON1, kind D, JSON body, and the XOR checksum *06, ending in a newline.

4.7.1 Frame Delimitation

Framing is by sentinel and not by length. Every microcontroller prints boot banners and debugging text, so a length-prefixed parser would resynchronise after such noise only by luck. A sentinel parser discards anything not beginning with BEACON1 as counted noise and continues, which makes the protocol self-synchronising: it recovers at the next newline after any corruption, with a loss of at most one frame.

4.7.2 Error Detection

The checksum is an 8-bit XOR over the body (a longitudinal redundancy check). It detects all single-bit errors, but it cannot detect an even number of bit errors in the same bit position, any reordering of bytes, or corruption of the sentinel and kind fields, which lie outside the body. Under a random-corruption model, 0.39% of corrupted frames would pass. A CRC-8 occupies the same eight bits and detects all bursts up to eight bits and byte transpositions, so it is recorded as the protocol's clearest improvement. It was not adopted for the first bring-up because the XOR form is three lines of C and legible in a serial monitor.

4.7.3 Channel Utilisation

Table 4.3 gives the measured frame sizes and the derived link performance. At 1 Hz the link is over-provisioned by more than two orders of magnitude, so the verbose JSON codec is the default: it costs 39% more bandwidth than the compact codec and buys human-legible frames during bring-up.

Table 4.3: Frame Size and Link Performance of the JSON Codec

Quantity	Value
Data frame, mean size	72.4 B
Bits on the wire per frame	724
Maximum sustainable rate at 115200 baud	159 frames/s
Utilisation at 1 Hz, 115200 baud	0.63%
Utilisation at 1 Hz, 9600 baud (as built)	about 7.5%

4.8 IMPLEMENTATION OF REFUSAL GATES

A request through the serial path passes the following stages. The route validates the request, a line source yields lines (decode errors become counted rejections, not

Table 4.6: Design Specification Versus Implemented Rig

Element	Design document	As built
Microcontroller	ESP32-WROOM-32 DevKit	NodeMCU ESP8266 (CH340 USB bridge)
Current and voltage	INA219 on I ² C	INA219
Temperature	DS18B20 (1-Wire)	LM35
Cell	Protected 18650	HONGLI ICR-18650, single cell
Serial rate	115200 baud	9600 baud (CH340 bridge limitation)

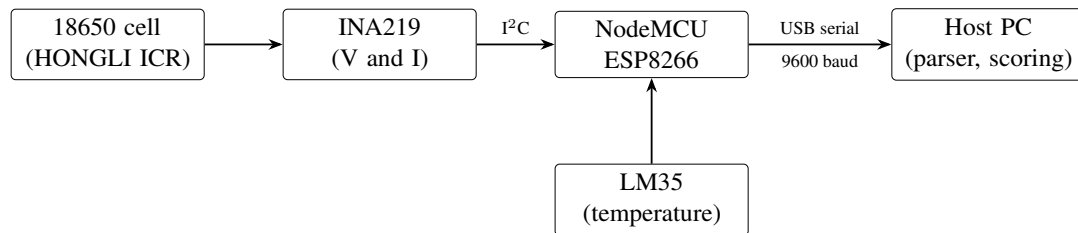


Figure 4.2: Implemented bench rig: measurement chain from the cell to the host.

electronics are powered from USB, not from the cell. The design power budget is about 48 mA typical against 500 mA available from USB, and the firmware keeps the radios disabled, not to save power but because transmit current spikes can brown out a weakly supplied board, which would restart the time base.

4.9.3 Firmware

The firmware is one board-agnostic Arduino sketch. The sensors sit behind four replaceable functions (`readVoltageV`, `readCurrentA`, `readTemperatureC` and `readSocPercent`), so a board with only some sensors can still run. At boot it probes each sensor, emits a status record naming what failed, and declares in its `HELLO` only the channels it can actually supply, together with the rated capacity. The host then scores the channels present and refuses the rest by name. The sketch compiles for three targets on every push (Table 4.7); the continuous-integration build is the stub configuration without the sensor libraries, and a test compares its declared field set, as a set, against the schema.

Table 4.7: Firmware Compilation Record (`arduino-cli`, all warnings enabled)

Target	Flash used	RAM used
Arduino Uno (AVR)	8,990 B of 32,256 (27%)	460 B of 2,048 (22%)
NodeMCU v2 (ESP8266)	240,340 B of 1,048,576 (22%)	28,372 B of 80,192 (35%)
ESP32 Dev Module	272,324 B of 1,310,720 (20%)	22,140 B of 327,680 (6%)

The `/healthz` endpoint is explicitly service liveness and not battery health: in a battery-monitoring system the naming collision is a real hazard and is called out at its definition. The container health check polls it.

The React client is a single-surface dashboard. It shows the fleet of batteries with health, risk and remaining life for each, a battery detail view, a bench-rig panel, a digital-twin sheet, and a banner stating where the run's data came from. An earlier version had separate tabs for fleet, CAN replay, thermal and transfer views. They were removed because two parallel dashboards meant two provenance banners that could disagree about the same run. The twin sheet is present but displays “not available” for the observed and predicted state, because the twin module does not yet emit its state into the dashboard payload, and no substitute trajectory is drawn. A Guardian explanation is shown next to the run it explains, and refusals are rendered as results and not as errors.

Fig. 4.3 and Fig. 4.4 show the dashboard. Both are screenshots of the running system, taken on a synthetic fleet, which the interface labels as simulated data; the bench-rig panel below the fleet summary is the only part backed by a physical measurement.

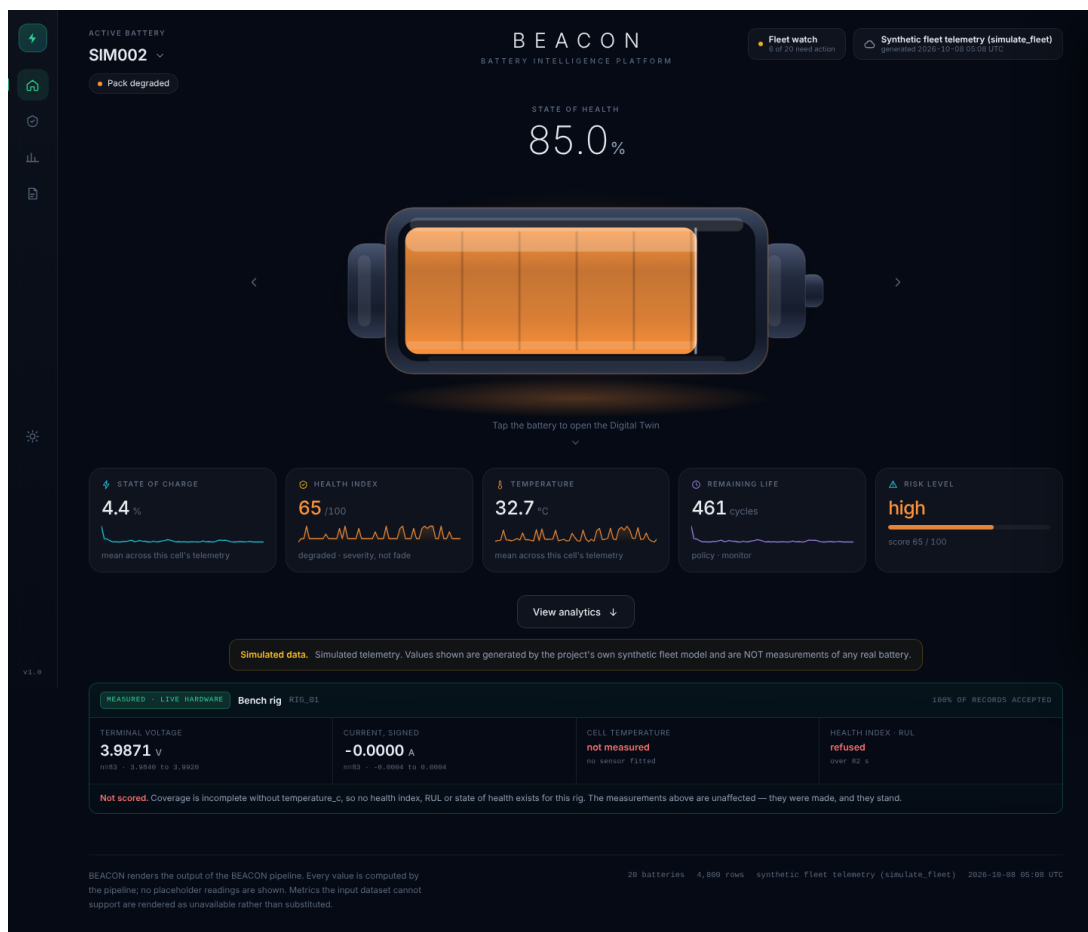


Figure 4.3: Overview screen of the BEACON dashboard on a synthetic fleet. The banner marks the values as simulated, and the panel at the bottom shows the measured bench-rig readings, with the health index and remaining life refused because no temperature channel exists.

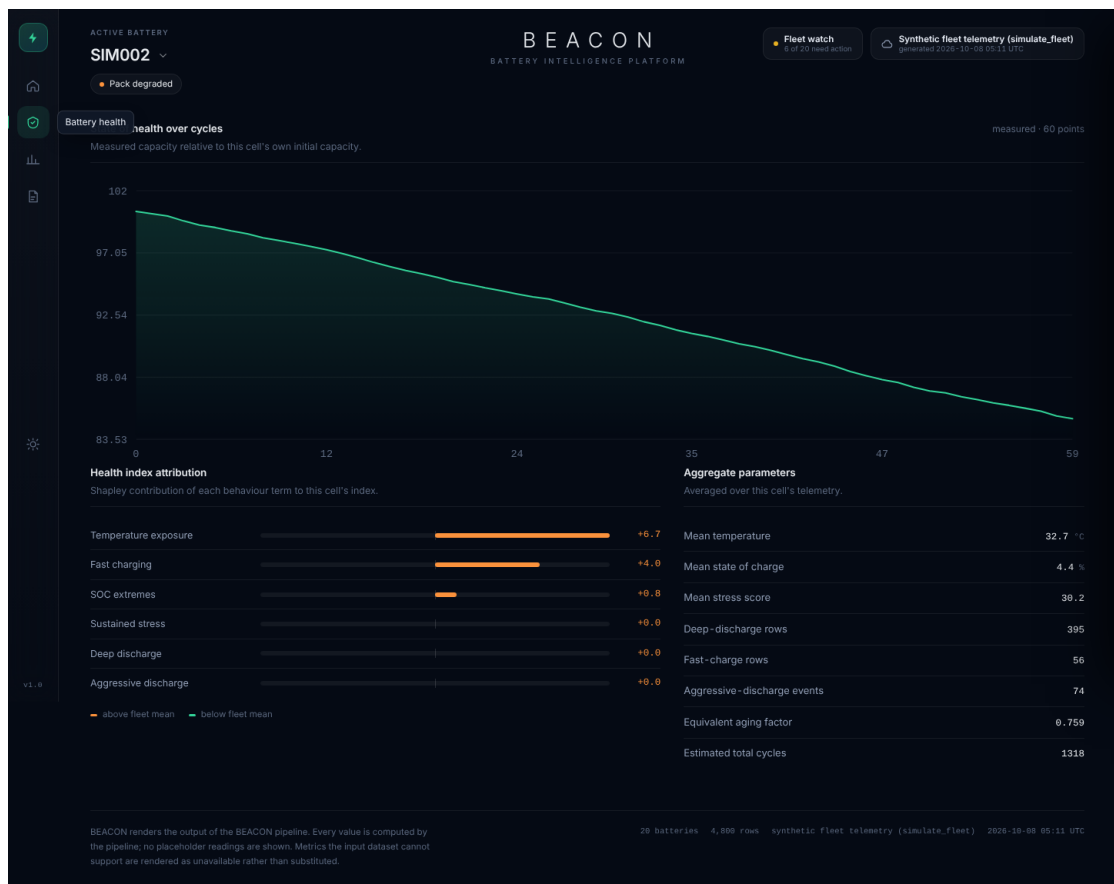


Figure 4.4: Battery health screen: state of health over cycles for one battery, the attribution of its health index to behaviour terms, and its aggregate usage parameters (synthetic fleet).

There are two deliberate absences. There is no database: the fleet store is in memory, and adding a database to the compose file would imply a persistence layer that does not exist, so the compose file says so. Persistence is the next substantial piece of work.

4.11 DIGITAL TWIN AND FLEET STORE

The twin is implemented as a domain module with no transport dependency. The API layer is a thin adapter over it, so adding another front end, for example a message broker, means writing a new adapter and leaves the module unchanged. The fleet store holds the latest twin state, the transition history and the telemetry of the most recent run for each battery. It is held in process memory and does not survive a restart, which is stated in the API documentation and shown in the compose file by the absence of a database service.

The twin also carries a field called evidence confidence, which takes one of four values: validated, heuristic, mixed or not applicable. It states whether the causes the Guardian identified for a battery are supported by the calibration work or are hand-chosen thresholds. Temperature exposure is the only cause with a measured, cohort-controlled effect (0.0038 Ah per °C), while the current-based causes and the health index itself are

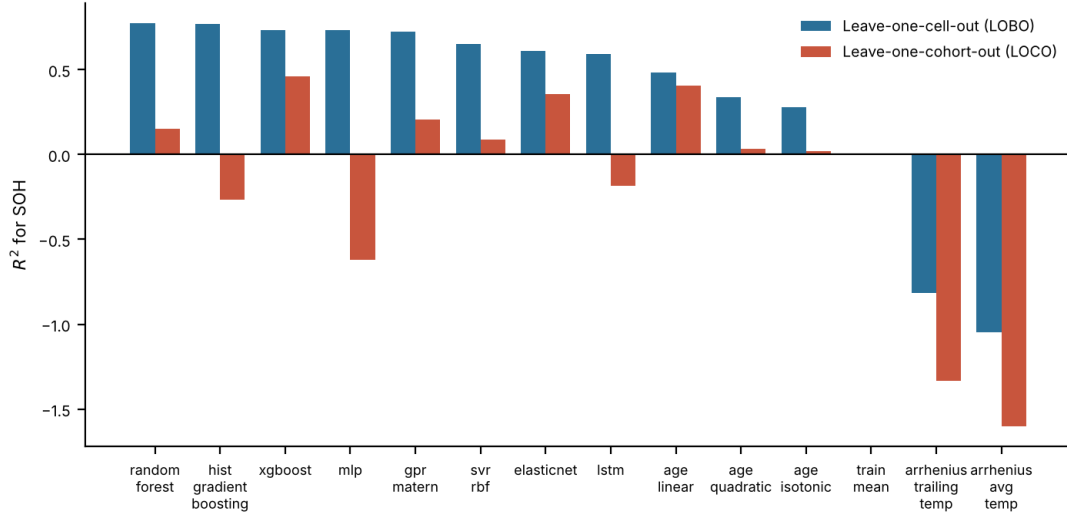


Figure 5.1: LOBO and LOCO R^2 for SOH for every scored method on the 31-cell NASA benchmark (data: `benchmark_results.csv`). Skill measured across cells does not carry over to a new protocol.

Table 5.2: Feature Ablation and Shuffle Control (NASA SOH, LOCO)

Run	Method	LOCO R^2	95% CI
All features (published)	XGBoost	0.459	$[-2.80, 0.86]$
All features (published)	Age-linear	0.406	$[-0.43, 0.51]$
Cycle number only	XGBoost	-0.266	$[-1.15, 0.82]$
Behaviour features only	XGBoost	-0.293	$[-0.72, 0.80]$
Shuffled in time (control)	Age-linear	-0.039	$[-0.35, -0.004]$
Shuffled in time (control)	XGBoost	-0.436	$[-1.44, -0.008]$

Temperature was the only usage factor with a consistent, correctly signed within-protocol relationship to fade (seven of seven cells in the two cohorts where it was significant), and its magnitude did not transfer between protocols. The rule-based risk score did not track measured fade either ($\rho = -0.27$, $p = 0.12$, on the 33 cells of the earlier health-index ranking), so it is labelled a heuristic.

Interpretation. No fitted method retains reliable skill under a change of protocol, and no method can be distinguished from another. This is the result that motivates the rest of the project: the estimators that ship fit nothing across cells.

5.2 FIELD STATE-OF-HEALTH ESTIMATION

Table 5.3 compares three ohmic treatments at the primary window on 22 CALCE cells. The field correction matched the laboratory correction within its interval and measured five more cells than no correction. Sensitivity windows gave 1.2% (4.05–3.55 V, 16 cells) and 1.9% (3.85–3.55 V, 17 cells).

The denominators differ because a correction can only be scored on cells for which

Table 5.3: Field SOH at the 3.90–3.60 V Window; Median Per-Cell Absolute Error Against Cyclers Capacity, 95% Bootstrap Interval Over Cells

Ohmic correction	Cells	Median	95% CI	Worst
None	12/22	4.3%	2.5–5.4%	6.4%
Cycler resistance (laboratory only)	12/20	1.4%	1.2–2.5%	5.6%
Step resistance (field)	17/21	1.7%	1.4–4.3%	10.3%

it exists: one cell never rests before discharging, so no step resistance exists for it, which gives 17 of 21 cells for the field arm and 17 of 22 once that cell is counted as refused. The three worst cells (6.4–10.3%) are all Type 3, whose discharge switches rate within a cycle. Every refusal carries a stated cause: four storage and partial-protocol cells delivered 80–155 equivalent full cycles before any constant-current discharge crossed the window, so the reference gate refused them, and one cell never rests before discharging. Fig. 5.2 tracks the estimate against every laboratory capacity test, and Fig. 5.3 shows every measured cell.

State of health from voltage, current and time only, against lab capacity tests

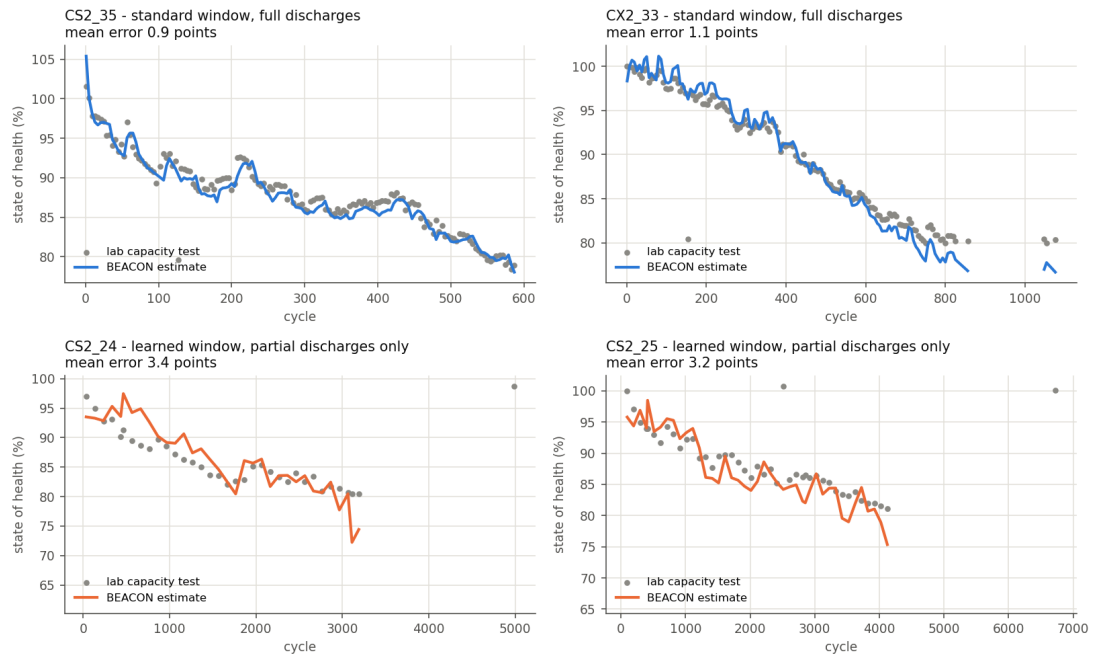


Figure 5.2: SOH from voltage, current and time against every laboratory capacity test. Top: two constant-current cells, fixed window (mean error 0.9 and 1.1 points). Bottom: two cells of real partial cycling, learned window, estimate from partial discharges only (3.4 and 3.2 points).

Interpretation. Identifying resistance from the load step took the median error from 4.3% to 1.7% and the coverage from 12 to 17 cells, without any quantity the cycler computed. Two earlier figures were corrected downward by this work: an earlier SOH figure and an 88% RUL figure had been scored against a reference capacity taken from each cell’s whole life, which uses the future and which no BMS can hold.

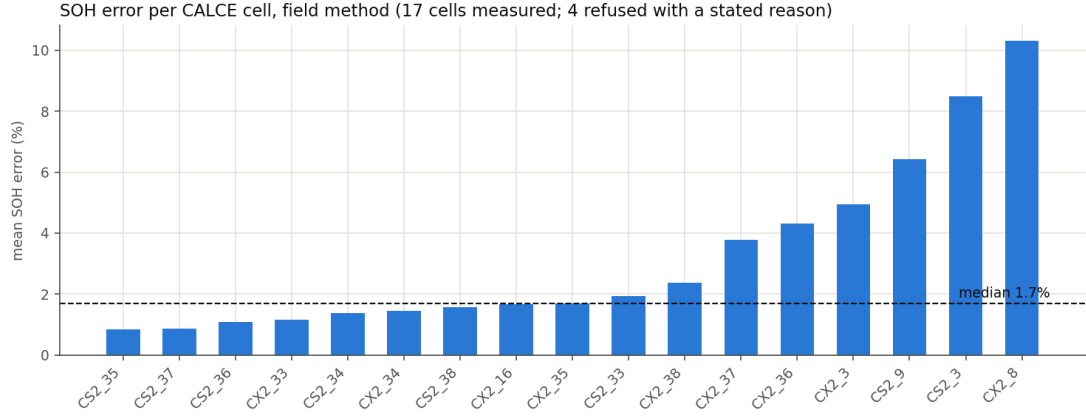


Figure 5.3: Mean absolute SOH error for every measured CALCE cell, field method; the dashed line is the median (1.7%).

5.3 PARTIAL-DISCHARGE SOH ESTIMATION

On the four cells of real partial cycling (Table 5.4), estimates use only the partial discharges preceding each laboratory check. Before the knee gate existed, the two near-empty cells were reported at 6.8% and 12.8% error while appearing confident. Learned windows on the 18 full-discharge cells have steepness 0.44–0.88, so any threshold between about 0.9 and 8 gives the same verdict on every cell and the chosen 2.0 is not tuned. As a control, the learned mode on those 18 cells measured 16 at a median of 1.95%, beside 1.7% for the fixed window, so it is a fallback and not a replacement.

Table 5.4: SOH from Real Partial Cycling (CALCE CS2 Types 5 and 6)

Cell	Band	Learned window	Steepness	Result
CS2_24	top of charge	4.00–3.82 V	0.83	3.4% error
CS2_25	top of charge	4.03–3.83 V	0.82	3.2% error
CS2_5	near empty	–	10.4	refused (knee)
CS2_6	near empty	–	8.2	refused (knee)

Interpretation. The refusal is physics and not a tuned cut. On the plateau, window charge tracks capacity, while on the knee voltage is set by resistance and diffusion, so it measures how hard the cell is working instead. The scope is two cells per band, which shows the mechanism and is too few to quote a population error.

5.4 INTERNAL RESISTANCE ESTIMATION

Against the cyclers’ internal-resistance column over each cell’s life (20 cells carrying both values), the step resistance achieved a median Spearman correlation of 0.86, growth in the same direction on 19 of 20 cells, and a median growth of 1.26 times against the cyclers’ 1.29 times. This includes the partial-cycling cells (0.90 and 0.98), and the exception is a Type 3 cell. The levels differ, since the step value includes early

polarisation, but the trends agree, which is what a power-fade indicator requires.

5.5 REMAINING USEFUL LIFE ESTIMATION

At threshold 0.90, referenced to each cell’s first cycles (17 cells), Table 5.5 reports accuracy by the true distance to end of life and by the prediction a user sees. Near end of life the estimate is a safe lower bound: it warns early and almost never late. Errors are predominantly early because these cells fade quickly at first and then more slowly, so a full-history line is steeper than the remaining fade.

Table 5.5: RUL to 90% Capacity. Left: By True Cycles to End of Life. Right: By Predicted Cycles, With the 90% Range of (True – Predicted)

True	Within ± 20	Median err.	Predicted	Lastest $\geq$ pred.	90% range
0–25	73%	9.6	0–25	95%	+1 to +82
25–50	44%	22.8	25–50	88%	–14 to +129
50–100	15%	42.7	50–100	85%	–40 to +141
200–400	0%	144	–	–	–

Scored against a reference capacity taken as the 95th percentile of the cell’s whole record, the same estimator was within ± 20 cycles 88% of the time near end of life. That reference uses the future, which no BMS can hold. Against the cell’s own early capacity the figure is 73%, and that is the figure reported. Fig. 5.4 summarises the result.

Remaining life to 90%: 73% within ± 20 cycles in the last 25 cycles (median miss 10 cycles)

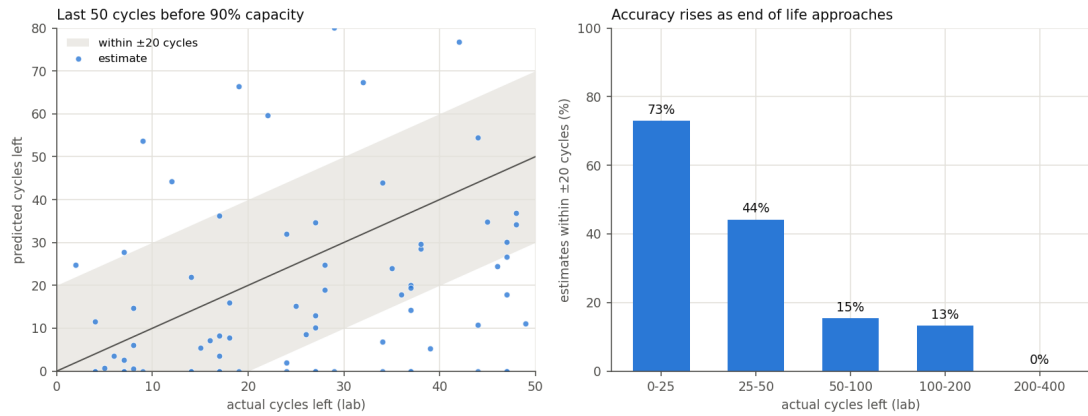


Figure 5.4: RUL to 90% capacity. Left: predicted against actual cycles left in the last 50 cycles. Right: share of estimates within ± 20 cycles by actual distance to end of life.

Interpretation. The “within ± 20 cycles” claim is true near end of life and false far from it (0% at 200 or more cycles out), so it is reported with its condition and not bare. The conservative direction is a property of the 90% threshold on these cells and is not a general guarantee. At an 85% threshold the median bias near end of life turns positive (+6.7 cycles within 25 cycles of end of life), so the estimate runs late there and only 62.8% of estimates fall within ± 20 cycles. The conventional 80% end-of-life

as a cause and which led to the fix.

Fig. 5.5 plots the Stage B capture and Fig. 5.6 shows how the dashboard presents it.

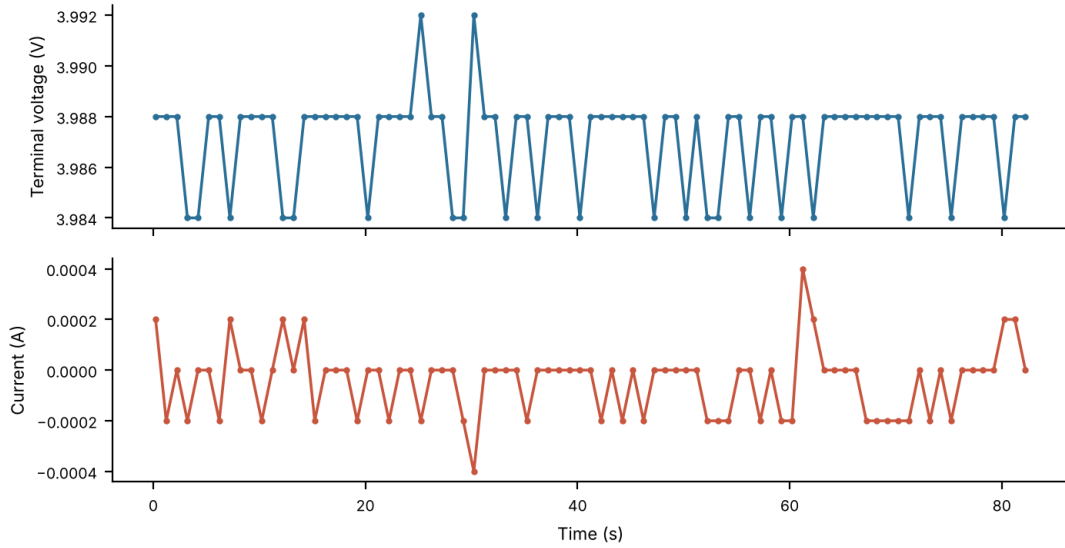


Figure 5.5: Stage B capture from the bench rig (83 frames over 82 s, one cell at rest). Voltage moves between 3.984 and 3.992 V in steps of the INA219’s 4 mV LSB, and current stays at the ± 0.4 mA noise floor.

5.10.2 Fault Injection

Faults injected into a real capture behaved as designed: a corrupted byte failed its checksum, a NaN field was rejected, a reversed timestamp refused the capture, and a missing sensor was refused by name. The injection also exposed one defect, since corrected: a 999 V reading passed the field range, which is wide so that packs are admissible. Records outside 0–5 V are now rejected when the source declares a single cell.

5.10.3 Scope of the Hardware Results

The captures demonstrate that the transport, the protocol, the gate and the refusal path work against real silicon. They are one cell, at rest, for under 90 s, with no load step and no charge or discharge cycle. They are *not* a validation of the health, RUL or risk models, which were fitted on cycled research cells and cannot be confirmed or refuted by a cell that was never cycled. Sensor accuracy against a reference meter was not measured, and no discharge has yet been recorded on the rig.

5.11 CASE STUDY: HEALTH REPORT CARD

Fig. 5.7 abridges the report card for CALCE cell CS2_35 as it would have read at cycle 90. The pipeline saw voltage, current and time only.

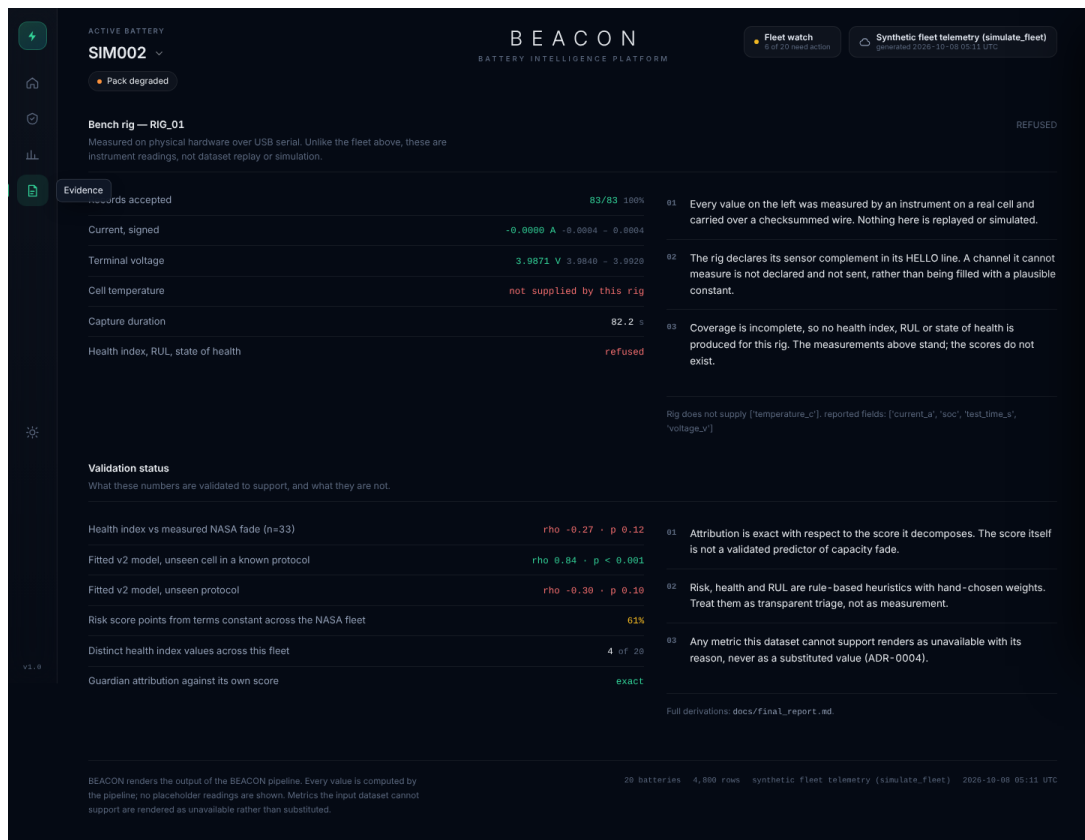


Figure 5.6: Evidence view of the BEACON dashboard for the bench rig: 83 of 83 records accepted, voltage and current as measured, temperature reported as not supplied, and the health index, remaining life and state of health refused because coverage is incomplete.

1. HOW HEALTHY IT IS
91.0% state of health [MEASURED] State: WARNING
2. HOW LONG IT HAS LEFT
About 5 cycles until it falls to 90% of its early capacity. [ESTIMATE]
3. WHAT YOUR USAGE IS DOING TO IT
No temperature was measured, so heat exposure cannot be assessed.
4. WHAT TO DO
Capacity fade is measurable. Keep monitoring.
5. WHAT THIS REPORT CANNOT TELL YOU
- Scoring skipped: required channels not supplied.

CHECK AGAINST THE LAB (the card was not shown this)
Lab capacity at cycle 90: 91.4%; card said 91.0%
Cell actually crossed 90% at cycle 146: 56 cycles after cycle 90; card said 5

Figure 5.7: Abridged health report card for CS2_35 at cycle 90, with the laboratory check.